Netaji Subhas University of Technology

Formerly NSIT, University of Delhi

Department of Computer Science & Engineering

Data Visualizations using R

MINI PROJECT REPORT

Unit II — Automation and Data Visualization

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GitHub Repository

github.com/TanishyadavNSUT/Data-Visualizations-using-R

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1 Dataset Architecture and Demographics

1.1 Introduction

This project leverages the power of R for comprehensive data visualization. Our focus is to analyze auditory dataset distributions, uncover hidden spectro-temporal interactions, and draw concrete analytical conclusions about feature viability in machine learning classification models. Data visualization provides the foundation for data-driven modeling, turning raw numerical arrays into interpretable patterns through geometrical mapping and colour aesthetics.

> **Dataset Definition 1.** The dataset comprises diverse acoustic recordings categorized broadly into foundational classes—such as coughing audio, persistent background ambiance, and miscellaneous human noise.

1.2 Class Distributions and Proportions

> **Visualization Context.** The following visualizations depict the quantitative makeup of the datasets prior to passing them through feature-extraction pipelines. Understanding the volume balances prevents biases. Bar charts are highly effective for univariate categorical data, translating raw counts into proportional lengths that the human eye can easily compare. By establishing these baseline frequencies, we ensure our machine learning models do not succumb to class imbalances.

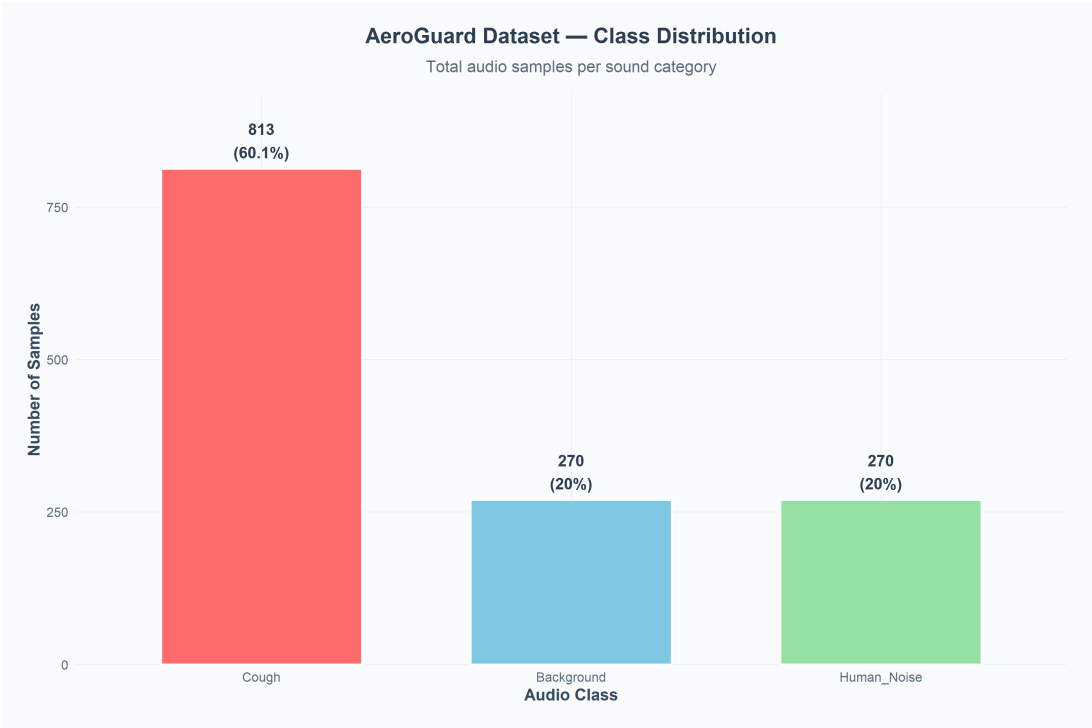

Figure 1: Absolute Class Frequencies (Bar Chart): This plot displays the gross frequency of available audio segments partitioned by respective behavioral classes. The raw count exposes whether our dataset favors background observations over target respiratory anomalies.

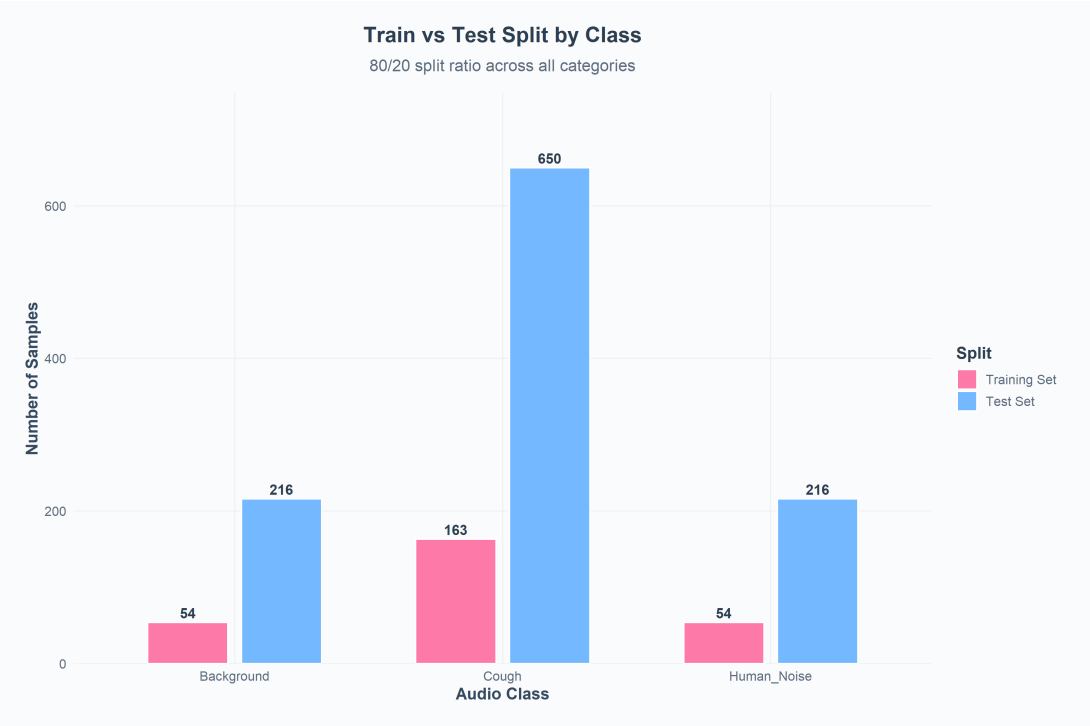

Figure 2: Train vs. Test Sampling Mapping: Illustrates how the distinct audio labels were fragmented into Training (validation) and Testing (inference) subsets. This helps confirm that real-world deployment accuracy represents a proportional amount of the data scale.

2 Spectro-Temporal Feature Analysis (MFCC)

2.1 Density and Feature Spread

> **Key Finding.** Mel-Frequency Cepstral Coefficients (MFCCs) accurately simulate human audio perception. Analyzing their distributions provides baseline limits used for optimal thresholding in on-device audio classification. Proper distribution analysis is critical for understanding the probability density functions of our input variables.

The plots in this section dive into the statistical range of specific coefficients. Box plots graphically depict groups of numerical data through their quartiles, offering an immediate grasp of the median, interquartile range, and outliers. Violin plots expand on this by layering a rotated kernel density plot on each side, conveying both internal summary statistics and exact probability densities at different values.

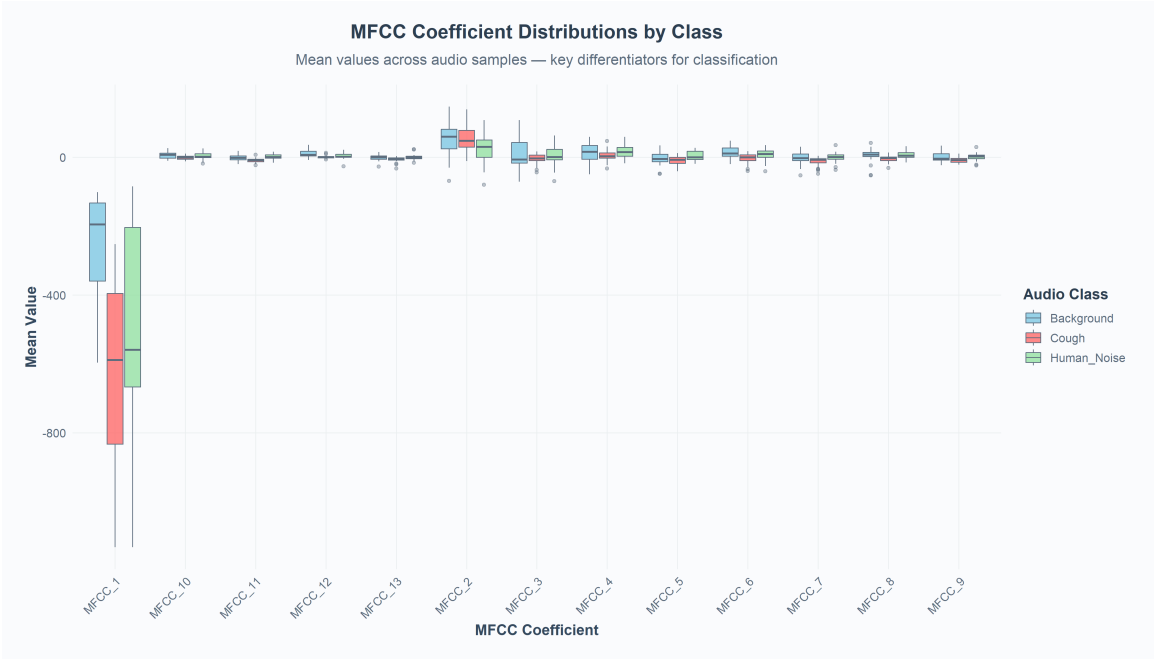

Figure 3: Feature Boxplots: Showcases the interquartile ranges, median values, and scattered extremes across multiple evaluated target classes. It clarifies which parameters possess massive variances versus strictly constrained limits.

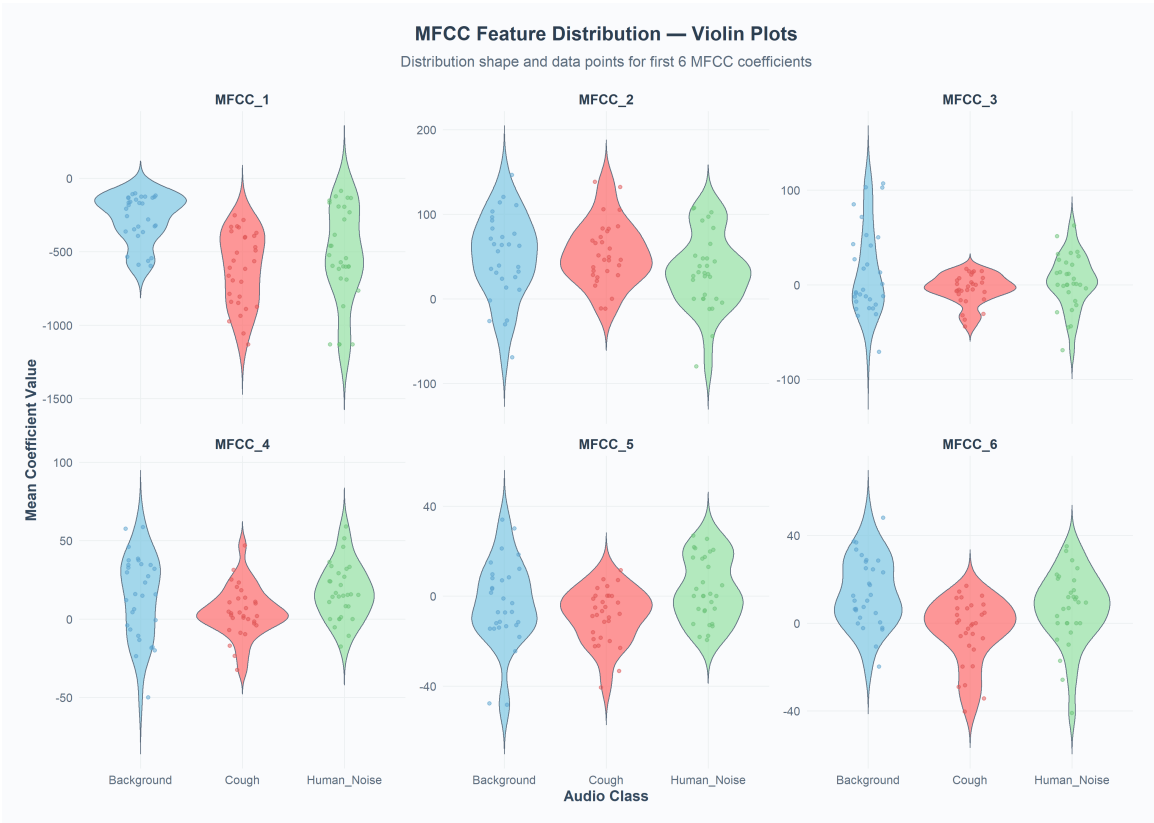

Figure 4: Density Violins: A more descriptive formulation than the boxplot, utilizing a mirrored kernel density estimation to reveal multi-modal peaks and true distributional structures of individual audio markers.

2.2 Correlations among Features

Understanding collinearity between multiple mathematical variables helps to optimize features by shedding redundant information and highlighting uncorrelated, entirely independent metrics. Correlation matrices represent normalized

covariance indices visually, utilizing heat scales to map strong linear dependences between sets of continuous variables.

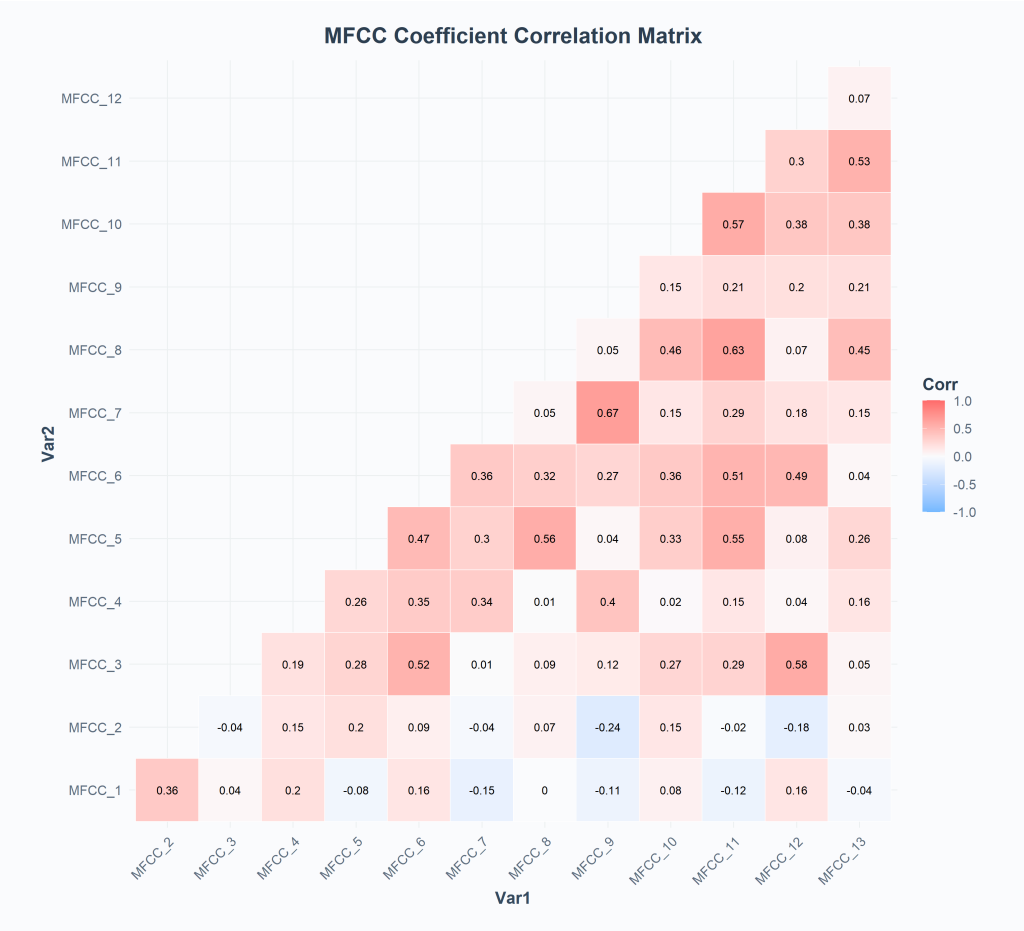

Figure 5: MFCC Correlation Matrix: Displays pairwise correlations among acoustic coefficients via color intensities. Lighter/darker regions signify heavy collinear interactions, effectively flagging pairs that scale uniformly together.

3 Granular Source Variability

Visualization Context. By isolating singular feature metrics—and charting source distributions alongside variable stability—we detect potential false-positive triggers originating from temporal shifts in background scenarios. Tracking variations over time via aggregate line profiles helps observe dynamic variance that singular histograms might obscure.

3.1 Profiling and Parameter Fluctuation

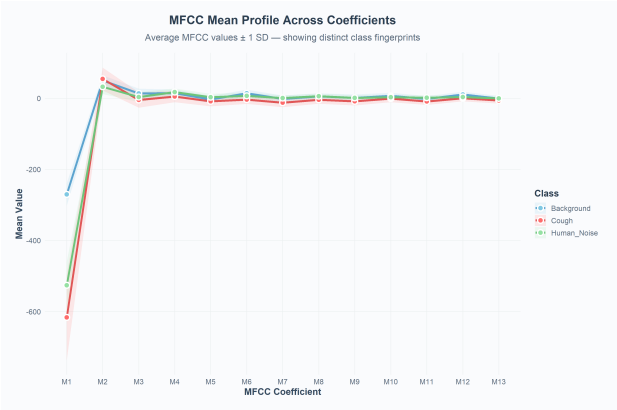

Figure 6: Aggregate Variable Profile: A comparative line trajectory plotting the average baseline value for features per distinct target class.

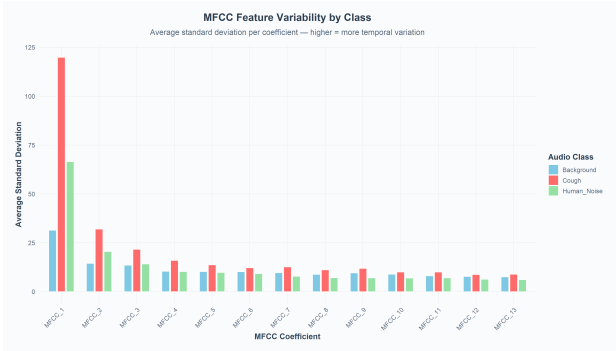

Figure 7: Variability Flux Metric: Determines structural variance boundaries by observing standard deviation behaviors recorded across sequential epochs.

3.2 Source Identities and Time Density

To account for real-world scenarios, datasets combine audio frames from multiple unique sources and map lengths into varied window frames. Histograms and density curves are vital here to map continuous observation intervals into cleanly discretized distribution bins.

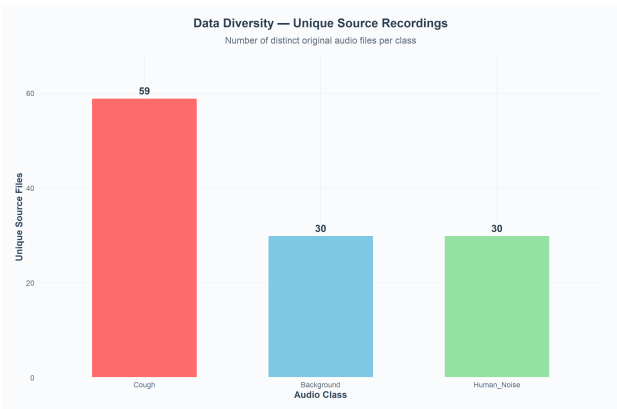

Figure 8: Unique Input Sources: Quantifies distinct contributor profiles providing acoustic captures.

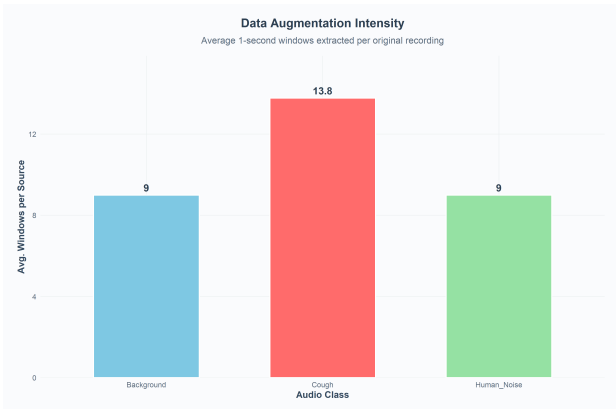

Figure 9: Captured Segments per Source: Highlights user-level variance in lengths contributed.

Figure 10: Temporal Window Density Overlap: Uncovers density distribution of the timeframe "window length" clusters, ensuring no single block over-influences the acoustic state sequence.

4 Embeddings and Advanced Summarization

> **Key Finding.** Mapping high-dimensional MFCC capabilities directly informs model success. Advanced visualizations like scatter maps, radar logic, and parallel coordinates evaluate inter-class decision boundaries comprehensively. These plotting algorithms project multidimensional geometry into comprehensible 2D formats, preserving vital structural separations.

4.1 Model Performance & Feature Geometry

Predictive model performance visualization distills error logs and evaluation scores into tangible lines of progression. Simultaneously, plotting continuous features against one another using scatter grids allows us to verify internal category clusterings.

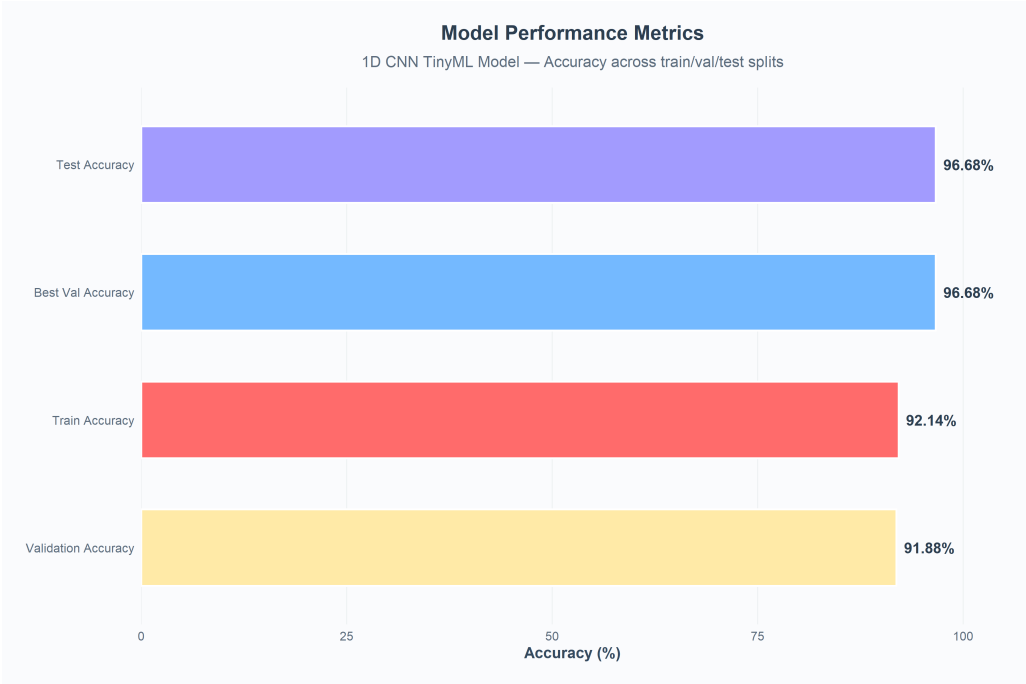

Figure 11: Algorithmic Accuracy Validation: A concise performance chart mapping prediction correctness rates directly against different benchmark iterations or configurations.

> **Observation.** Scatter distributions define intrinsic separability. If raw variable plots cleanly separate different behavior classes prior to neural network injection, real-world deployment viability increases substantially.

Figure 12: 2D Direct Scatter Mapping: Directly compares two primary features. The clustering properties reflect organic separability amongst competing acoustic triggers.

Figure 13: Comprehensive Feature Scatter Matrix: A pair-plot array mapping multiple variables concurrently. Shows multidimensional distribution patterns to track how variable intersections group the classes efficiently.

4.2 Specialized Multivariate Visualizations

Specialized graphs like Stacked Area maps provide insights into total volumetric changes of variables over time. Radar charts map multivariate data onto radial spokes to compare symmetry profiles. Parallel coordinates map multi-feature trajectories, treating mathematical categories as sequential axes to draw connecting behavior lines.

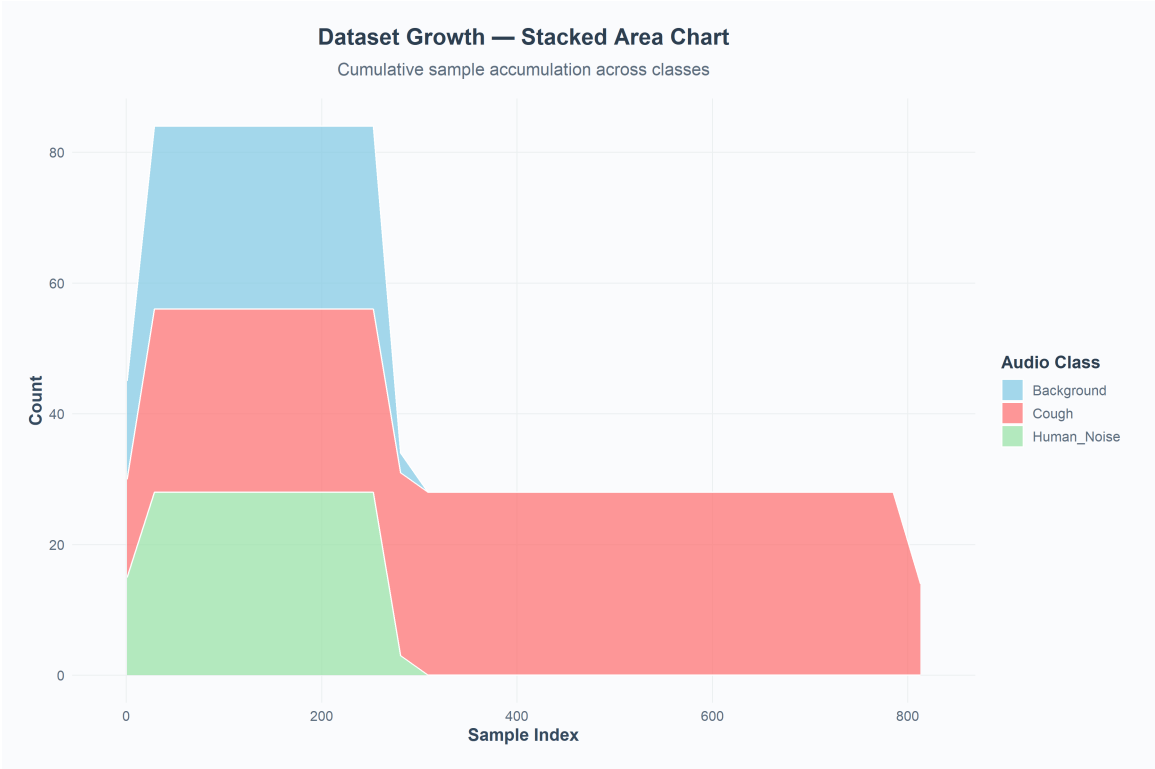

Figure 14: Stacked Area Density Map: Effectively cascades cumulative variable magnitudes. It reveals proportional feature shifts stacked on top of one another collectively relative to overall composition intensity.

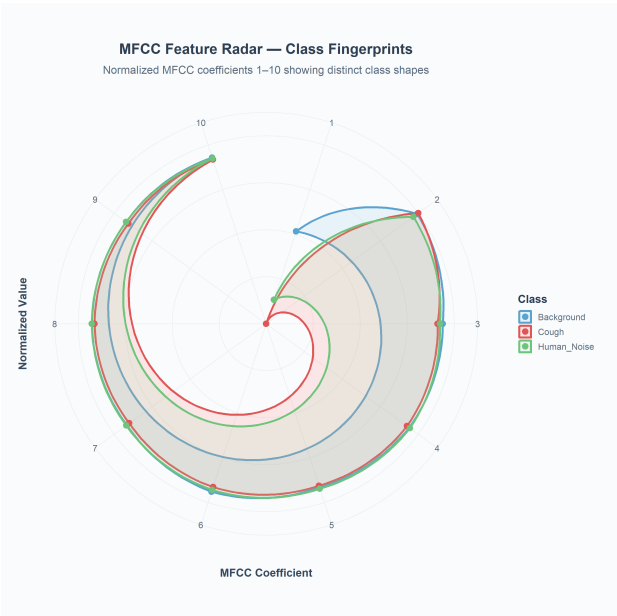

Figure 15: Radar Spatial Chart: Maps multivariate symmetry and spatial bias circularly along different variable spokes.

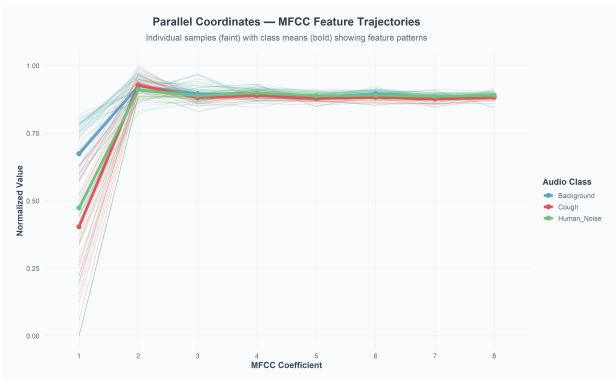

Figure 16: Parallel Coordinate Flow: Lines represent individual paths taken per observation traversing all feature columns.

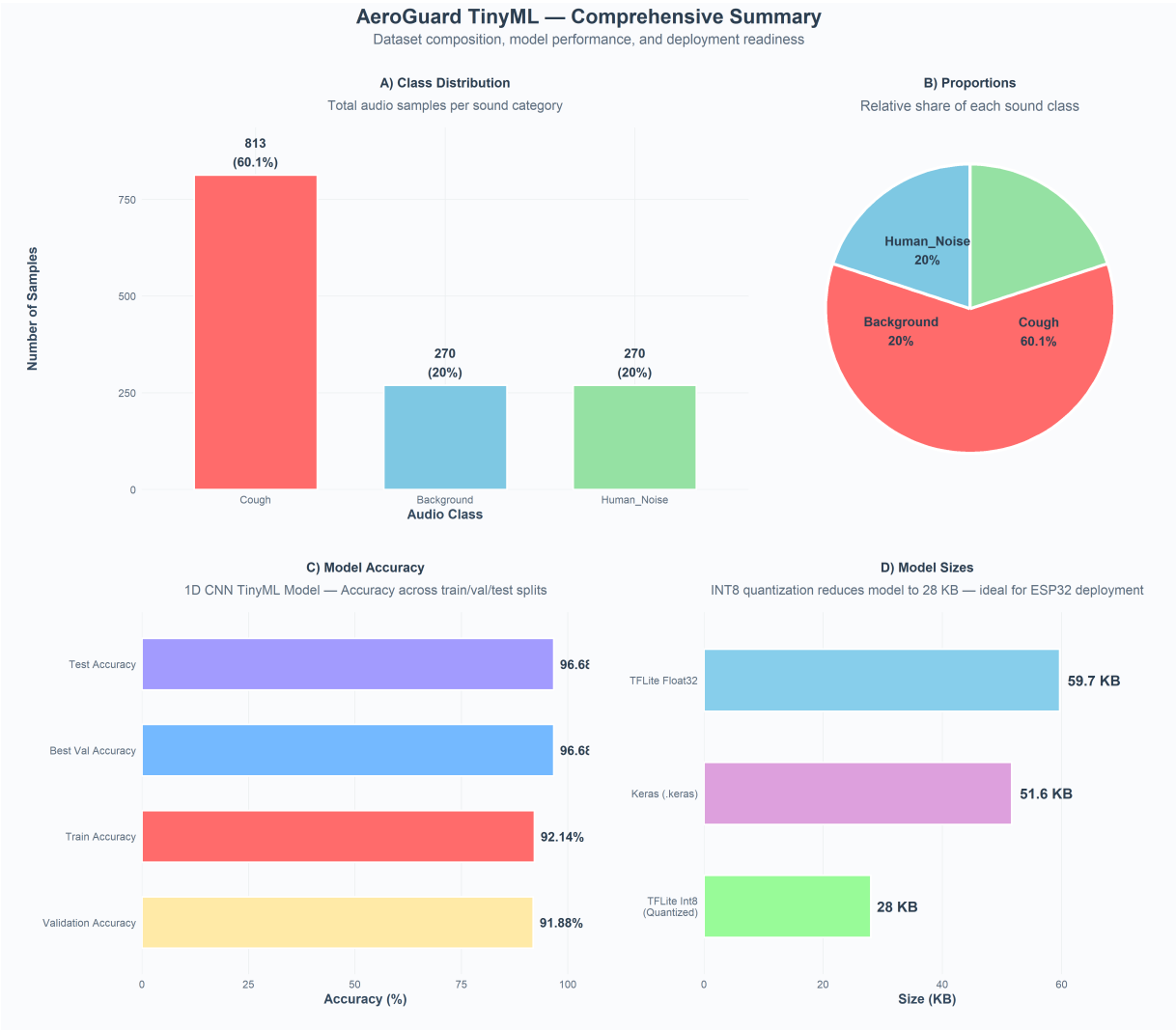

Figure 17: Comprehensive Dashboard Survey: An overarching composition layout summarizing key metric thresholds and final algorithmic balances inferred from our entirely R-driven data processing pipeline.